

Maria Rosaria Briglia

PhD student in AI for Government and Public Security, Sapienza University of Rome

Visiting researcher at NSF AI Institute for Foundations of Machine Learning (IFML), UT Austin
with Prof. Adam Klivans — November 2025 – present

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Research

Adversarial training through an energy lens

Reinterpreting robust classifiers as energy-based models, which reframes catastrophic overfitting and robust overfitting as behaviour of the energy landscape rather than isolated training pathologies — and turns that reading into concrete regularizers and sample weighting schemes.

The geometry of representation space

Concept control and prompt safety depend on the shape of the space you are steering in. Euclidean adjustments to text embeddings are the standard tool; hyperbolic space, with parallel transport in place of vector arithmetic, gives more expressive and more stable manipulation of concepts.

Threats to generative media, and defenses that hold

Proactive image protection rests on an assumed threat model, and I test those assumptions — whether watermark-style defenses survive diffusion-based reconstruction that re-synthesises an image while preserving how it looks. The broader question is what a defense is actually promising once the attacker has generative models of their own.

Generative models for proteins

On the structure side, benchmarking antibody–antigen complex prediction at scale: cofolding thousands of complexes to ask not only whether models recover the right interface, but whether their own confidence scores can tell which prediction did. On the design side, decoding-time methods for protein engineering that leave the base model untouched. A model that ranks its own outputs badly is the same problem whether the output is an image or a fold.

Publications

2026	Neutralizing Proactive Defense using Diffusion-based Upsampling <i>ACM Workshop on Information Hiding and Multimedia Security (IH&MMSec;)</i>
2026	Harnessing Hyperbolic Geometry for Harmful Prompt Detection and Sanitization <i>ICLR 2026</i>
2026	Semantic Steering via Hyperbolic Geometry <i>Beyond Euclidean Workshop, ECCV 2026 — Oral</i>
2026	Folding scFv–Antigen Complexes at Scale <i>GEM Workshop, ICLR 2026</i>
2025	What is Adversarial Training for Diffusion Models? <i>ICLR 2026</i>
2025	Implicit Inversion turns CLIP into a Decoder <i>ICLR 2026</i>
2025	Understanding Adversarial Training with Energy-based Models <i>arXiv preprint</i>
2024	Shedding More Light on Robust Classifiers under the Lens of Energy-based Models <i>ECCV 2024</i>

Education and positions

2025 Nov – present	Visiting researcher <i>NSF AI Institute for Foundations of Machine Learning (IFML), UT Austin</i> Visiting period under the supervision of Prof. Adam Klivans, applying representation learning and mechanistic interpretability to protein engineering.
2023 – present	PhD, National PhD programme in AI for Government and Public Security <i>Sapienza University of Rome</i> Developing practical defense techniques against the media manipulation enabled by generative AI.
2021 – 2023	M.Sc. Computer Engineering (LM-32), Artificial Intelligence and Intelligent Robotics <i>University of Salerno</i> Thesis: Automatic Generation of Identity-Preserving Gesture Videos.
2023	Research scholarship, artificial vision for autonomous driving on rail <i>RFI and University of Salerno</i> Designed the on-board hardware acquisition system, and the real-time anomaly detection along the railway track.
2018 – 2021	B.Sc. Computer Engineering (L-8) <i>University of Salerno</i> Graduated 110/110 cum laude, and recognised for completing the degree with highest honours in the shortest time possible. Thesis: Automatic Speaker Recognition Using a Neural Network Integrated into a Web Portal.

Summer schools

2025	CISPA – ELLIS Summer School, Saarbrücken, Germany
2024	Generative Modeling Summer School, Eindhoven University of Technology

Awards

2023	Recognition for an outstanding academic career, University of Salerno — for completing the B.Sc. with highest honours in the shortest time possible.
2023	Selected for courses reserved for excellent students, University of Salerno — open only to students averaging at least 28.

Technical

Python, C, Java, R, Matlab · PyTorch, TensorFlow · ROS Noetic, Django · Linux
Italian (native) · English (C1, Cambridge Advanced) · French (beginner)